VIT
CHENNAI

SUMMER SCHOOL ON MATHEMATICAL METHODS IN IMAGE PROCESSING

M2IP 2027

COHORT 1: IMAGE SEGMENTATION

Understand the mathematics. Test the method. Explain the result.

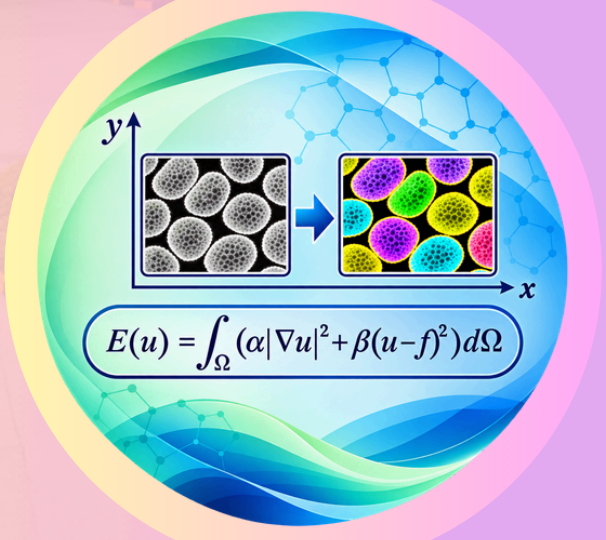
10-20 MAY 2027

(PHYSICAL MODE)

40 Participants

8 Project Mentors

27 Lecture Hours



A focused beginning in imaging research

Connect mathematical models with working algorithms. Learn through guided derivations, numerical exercises and a continuing team investigation into image segmentation.

Visit us at: https://m2ip.onrender.com

ORGANIZED BY

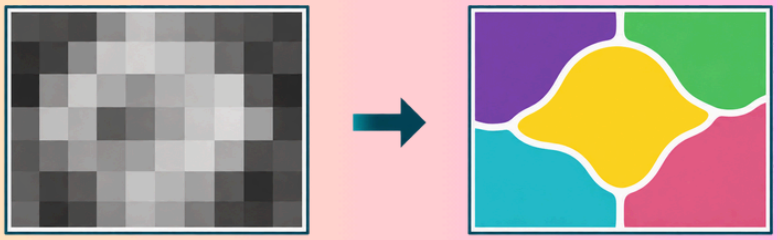
Department of Mathematics
School of Advanced Sciences
Vellore Institute of Technology
Chennai - 600 127

VIT - A PLACE TO LEARN, A CHANCE TO GROW

Applications Open
1st December 2026

Mathematics behind meaningful images

Image segmentation separates an image into regions that can be analyzed and interpreted. This school introduces early-career researchers to the mathematical ideas behind these methods and the practical decisions that influence their results.



WHY THIS SCHOOL?

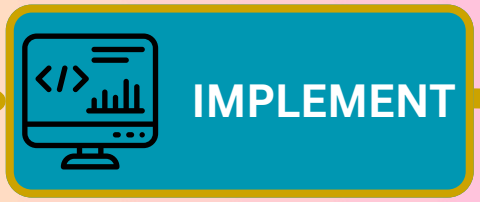
For mathematicians new to image processing, the central challenge is connecting an objective function or model assumption to a working algorithm. M2IP 2027 makes that connection through guided derivations, numerical exercises, and closely mentored experiments. Participants will study a focused set of methods in depth and examine when, why, and how their results change.

FOUR OBJECTIVES

01	Connect mathematical formulations with numerical implementation.
02	Investigate the effects of assumptions, noise, and parameter choices.
03	Enable six teams to complete focused comparative investigations.
04	Develop research skills through accessible training and a 30-day mentored follow-up.

HOW YOU WILL LEARN







Research begins with a precise question and an honest evaluation.

ABOUT VIT

Vellore Institute of Technology (VIT) was founded in 1984 as Vellore Engineering College, a self-financing inistitution dedicated to providing quality higher education. In 2001, the Union Ministry of Human Resource Development conferred university status on the institution under Section 3 of the University Grants Commission (UGC) Act, 1956. VIT is led by its visionary founder and Chancellor, Dr. G. Viswanathan, a former Parliamentarian and Minister in the Government of Tamil Nadu. In recognition of his exceptional contributions to education, Dr. G. Viswanathan was awarded an honorary doctorate by West Virginia University (USA) in 2009 and the State University of New York (USA) in 2024 With a commitment to delivering world-class education on par with international standards, VIT continually adopts innovative approaches to teaching and research. Our Memeranda of Understanding with various international universities, being a major strength, enable meaningful collaborations via student and faculty exchanges and foster joint research projects. Many of our students, working in international organizations and universities, deliver high-quality solutions and conduct research that bring esteem to India and make us proud of our legacy. At VIT, the pursuit of excellence in education and research is complemented by a dynamic ecosystem of innovation, research and global engagement. As we advance steadily, we remain steadfast in our mission to mature leaders and innovators who can make a meaningful impact on the world.

ABOUT VIT CHENNAI

VIT Chennai was established in 2010 and has since become a beacon of excellence in higher education. Spearheaded by Vice-President Dr. G. V. Selvam, a pioneering visionary who laid the foundation of VIT Chennai, he is an insightful architect of its evergreen campus and an ingenious leader whose comprehensive vision and mission have shaped its growth. Under his guidance, VIT Chennai has rapidly evolved into a hub of innovation and academic distinction. The leadership team, including Vice-Chancellor Dr. V. S. Kanchana Bhaaskaran, Pro-Vice Chancellor Dr. T. Thyagarajan, Director Dr. K. Sathiyanarayanan and Additional Registrar Dr. P. K. Manoharan, drives its mission of delivering transformative education and impactful research. Strategically located in the capital city of Tamil Nadu, VIT Chennai is a globally engaged, competitive, and research-enriched institution. VIT Chennai proactively addresses industrial, social, economic and environmental challenges, fostering application-based learning that produces industry-ready professionals. Known for its cosmopolitan atmosphere, VIT Chennai attracts students from across the globe, fostering a vibrant and multicultural community. Committed to maximizing industrial connectivity and establishing Centers of Excellence in cutting-edge research areas, VIT Chennai enriches technological and managerial human capital through a collaborative and inclusive academic environment. It serves as a platform for intellectual exchange, encouraging the convergence of diverse ideas to drive innovation and learning. Dedicated to societal and industrial advancement, VIT Chennai leverages its resources to benefit the community, supports national knowledge-building initiatives and fosters international collaborations for mutual progress in research and education. With a clear vision and an unwavering mission, VIT Chennai continues to redefine the landscape of higher education in India and beyond.

ABOUT SAS

The School of Advanced Sciences (SAS) at VIT Chennai, comprises Department of Mathematics, Physics and Chemistry. It offers the following programmes:

- Ph.D .
- M.Sc Data Science, M.Sc Physics, M.Sc Chemistry

The school also offers a plethora of core and elective courses facilitating B.Tech, M.Tech, MCA, Integrated M.Tech, Management and Law programmes.

COMMITTEE MEMBERS

CHIEF PATRON

Dr. G. Viswanathan
Founder & Chancellor, VIT

PATRONS

Dr. Sankar Viswanathan
Vice President, VIT

Dr. Sekar Viswanathan
Vice President, VIT

Dr. G.V. Selvam
Vice President, VIT

Dr. V. S. Kanchana Bhaaskaran
Vice Chancellor, VIT

Dr. T. Thyagarajan
Pro-Vice Chancellor, VIT Chennai

Dr. P. K. Manoharan
Additional Registrar, VIT Chennai

CONVENOR

Dr. N. Ramesh Babu
Assistant Professor, VIT Chennai

CO-CONVENORS

Dr. Lakshmanan Shanmugam
Professor, VIT Chennai

Dr. N. Padmaja
Assistant Professor, VIT Chennai

Dr. R. Sriraman
Assistant Professor (Sr. Grade), VIT Chennai

ORGANIZING MEMBERS

Faculty Members
Department of Mathematics - SAS, VIT Chennai

ADVISORY COMMITTEE

Dr. G. Vinitha
Professor and Dean, School of Advanced Sciences, VIT Chennai

Dr. A. Manivannan
Associate Professor and Head Department of Mathematics - SAS, VIT Chennai

SCIENTIFIC COMMITTEE

Dr. Mini Ghosh
Professor - HAG, VIT Chennai

Dr. Kalyani Desikan
Professor - HAG, VIT Chennai

Dr. Prabhakar V
Professor, VIT Chennai

Dr. Kaliyappan M
Professor, VIT Chennai

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M2IP 2027 - INVITED SPEAKERS



Ms. Afra Navas
Quantum Educator,
QClairvoyance Quantum Labs Pvt. Ltd., Hyderabad



Dr. Deepak Joshi
Associate Professor,
Centre for Biomedical Engineering, IIT, Delhi



Dr. Jayakumar Vaithiyashankar
Founder & CEO,
IBM Educator, Anuthantra Private Limited, Bengaluru



Dr. Karthik C
Assistant Professor,
School of Interdisciplinary Design & Innovation, IIITDM, Kancheepuram, Tamil Nadu



Dr. Karthikeyan Damodharan
Director,
Department of Vascular & Interventional Radiology, MIOT International, Chennai



Dr. Manas Ranjan Prusty
Associate Professor,
School of Computer Science & Engineering, VIT Chennai



Dr. Neha Dahiya
Scientist D,
Division of Implementation Research, Indian Council of Medical Research (ICMR), New Delhi



Dr. Nitika C. Panakkal
Associate Professor,
Department of Medical Imaging Technology, Manipal Academy of Higher Education, Karnataka



Mr. Pon Maa Kishan A
Founder & CEO of Sukish,
Technology Transfer Officer,
Department of Intellectual Property Strategy,
VIT Chennai

M2IP 2027 - PROJECT MENTORS



Dr. Priyanka Kokil
Associate Professor,
Department of Electronics & Communication Engineering,
IIITDM, Kancheepuram,
Tamil Nadu



Dr. Ramya Natarajan
Oncopathologist & Head
Department of Histopathology & Cytology, Neuberg Diagnostics Private Limited, Chennai



Mr. Sabarivel *R*
Regulatory Affairs Associate,
Tamil Nadu



Dr. S Sountharrajan
Professor ,
Department of Computer Science and Engineering,
Amrita School of Computing,
Amrita Vishwa Vidyapeetham,
Chennai Campus



Dr. Tarun Kumar
Project Engineer,
Quantum Technologies Lab,
Software Technology Division (STD),
Centre for Development of Advanced Computing (C-DAC), Mohali, Punjab



Dr. Taruna Madan Gupta
Scientist G & Head,
Division of Development Research,
Indian Council of Medical Research (ICMR), New Delhi



Dr. Vikram Patil
Professor & Vice Principal,
Department of Radio Diagnosis,
JSS Medical College,
Mysuru, Karnataka

SYLLABUS (M2IP 2027)

27 lecture hours | **Nine teaching mornings**

A focused progression from mathematical foundations to reproducible segmentation experiments.

Module	TOPICS	Hours
01	**MATHEMATICAL FOUNDATIONS** Images as discrete functions. Sampling and quantisation. Pixel neighbourhoods, convolution, smoothing, and gradients. Segmentation as a partition of the image domain. ***Practice:** Represent an image numerically and examine changes produced by simple filters.*	03
02	**CLASSICAL SEGMENTATION** Global thresholding and Otsu's criterion. Region growing. Reference masks and baseline evaluation. ***Practice:** Implement a baseline and inspect its failure cases.*	03
03	**CLUSTERING AND OPTIMISATION** K-means objective and alternative updates. Initialisation and convergence. Fuzzy C-means, constrained optimisation, membership updates, and the fuzzifier. ***Practice:** Derive updates and compare hard and fuzzy partitions.*	06
04	**SPATIAL MODELS AND REGULARISATION** Intensity fidelity and neighbourhood disagreement penalties. Two-label energies. Single-pixel label updates. Effects of regularisation on boundaries and small regions. ***Practice:** Implement a simple spatial model and vary its penalty.*	06
05	**EXPERIMENTAL DESIGN AND VALIDATION** Research questions, controlled comparisons, and fair baselines. Noise and parameter sensitivity. Dice, IoU, and boundary error. Separate tuning from testing. Repeated trails and variability. ***Practice:** Design and document a focused comparative experiment.*	06
06	**REPRODUCIBLE RESEARCH** Dataset provenance, code organisation, settings, and random seeds. Mathematical exposition, results interpretation, limitations, and research continuation. ***Practice:** Prepare a reproducible report with figures and tables.*	03

PROGRAMME & RESEARCH OUTCOMES

DAILY SCHEDULE (10-15 & 17-19 2027)	
09:30 - 11:00	Lecture I
11:00 - 11:15	Tea Break
11:15 - 12:45	Lecture II
12:45 - 13:45	Lunch
13:45 - 14:45	Parallel Team Monitoring
14:45 - 15:00	Tea Break
15:00 - 16:30	Guided Project Work

16 May 2027
No scheduled sessions.

20 May 2027

- Project presentations
- Expert feedback
- valedictory
- No lectures

ILLUSTRATIVE PROJECT QUESTIONS

- How does noise change the performance of K-means and fuzzy C-means?
- When does spatial regularisation remove small but meaningful regions?
- How sensitive is a segmentation result to initialisation or parameter choice?

Each candidate selects one manageable question with their mentor.

EXPECTED OUTPUT

- A precise problem statement and mathematical formulation.
- Documented code and reproducible experimental settings.
- An explanation of failure cases and limitations.
- A realistic research continuation plan.

AFTER THE SCHOOL

- **Within 15 days:** Submit interim results for mentor feedback.
- **Within 30 days:** Submit the refined report and code.
- **Dissemination:** Selected reports may be developed into book chapters after further revision. Publication depends on editorial review and publisher acceptance.

BEST RESEARCH PROJECT AWARD

- Presented to the team, demonstrating mathematical clarity, sound implementation, rigorous evaluation, and reproducible results.
- Selection will follow assessment of the final project report and code submitted within 30 days after the school.
- Well-supported negative findings will receive equal consideration.

APPLICATIONS & PARTICIPATION

WHO CAN APPLY?

- Postgraduate students, research scholars, postdoctoral researchers and early-career faculty in mathematics or related disciplines.
- Applicants should have basic knowledge of linear algebra, calculus, and probability, and be able to write simple Python/MATLAB programs.
- Prior research experience in image processing is not required.
- Participants must commit to the full school, team assignments, and the 30-day project follow-up

HOW WILL CANDIDATES BE SELECTED?

- Selection will be based on academic merit and a statement of interest of 150–200 words, outlining your academic background, research interests, reasons for attending, and expected benefits.
- Preference will be given to postgraduate students, research scholars, and early-career researchers below 35 years of age.
- Among equally suitable applicants, preference will be given to those with limited access to similar training, considering institutional and geographical diversity.
- 40 participants will be selected, with a reserve list maintained to fill vacant seats.

IMPORTANT DATES

Applications open	01 December 2026
Application deadline	30 December 2026
Selection list announced	03 January 2027
Registration and fee payment deadline for selected candidates	15 March 2027
Summer School	10-20 May 2027

FEE REFUND & TRAVEL SUPPORT

- Participants with at least 90% attendance who complete all prescribed assessments will be eligible for a fee refund, subject to funding availability.
- Travel support will be subject to funding availability, approved limits, and submission of valid travel documents.
- Refund conditions and travel support limits will be communicated to selected candidates before fee payment.

REGISTRATION FEE

Rs. 2000 per selected participant	The selected candidate should confirm participation and pay the fee by 15 March 2027

ACCOMMODATION

Accommodation availability and applicable charges will be communicated in the selection notice. Accommodation may be provided free of charge, subject to funding availability.

SCAN FOR APPLICATION

WEBSITE

HOW TO REACH